POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 5031 | Semester V | Theory | 4 Credits

Electrical Machines II

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electrical & Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 5031 as Electrical Machines II in Semester V for Electrical & Electronics Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Kharagpur’s Electrical Machines - II course and polytechnic electrical machine curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Machine designs, testing procedures, grid-synchronization protocols and maintenance plans must be based on approved manufacturer data, certified equipment testing, current safety standards and competent professional review.

Verified source note: Verified sources support rotating magnetic field, induction motors (3-phase/1-phase), alternators, synchronous motors, voltage regulation and parallel operation. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Electrical Machines II studies the principles, construction and performance of alternating current (AC) rotating machines. It develops the ability to analyze three-phase and single-phase induction motors, synchronous generators (alternators) and synchronous motors, evaluate their operating characteristics, starting methods and speed control while respecting the technical and safety boundaries of electrical machine operation.

> **Malayalam support:** □ handbook □□□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□□ □□□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Basic electrical engineering, electrical machines I (DC machines and transformers), electrical circuit theory and engineering mathematics.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the principles, construction and performance characteristics of three-phase induction motors.
2. Explain the construction, EMF equation, voltage regulation and parallel operation of synchronous generators (alternators).
3. Explain the working principle, starting methods and performance characteristics (V-curves) of synchronous motors.
4. Explain the types, working principles and applications of single-phase induction motors and special AC machines.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□□ exam-□ □□□□□ □□□□□□□ □□□□□□ □□□□□ □□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain the construction, rotating magnetic field and torque-slip characteristics of three-phase induction motors.	Understand
CO2	Explain the construction, voltage regulation and parallel operation of alternators.	Understand
CO3	Explain the working principle and performance analysis of synchronous motors.	Understand
CO4	Explain the working principles of single-phase motors and special electrical machines.	Understand

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

Module	Title	Official major topics	Hours	CO	Handbook section
1	Three-Phase Induction Motors	Constructional details (squirrel cage and slip ring); production of rotating magnetic field; working principle; slip and its significance; torque-slip characteristics; equivalent circuit; starting and speed control; braking.	Approx. 16 hours	CO1	Module 1
2	Synchronous Generators (Alternators)	Constructional features (salient pole and cylindrical rotor); EMF equation; winding factors; armature reaction; synchronous reactance; phasor diagram; voltage regulation (EMF, MMF methods); parallel operation.	Approx. 16 hours	CO2	Module 2
3	Synchronous Motors	Working principle (magnetic locking); starting methods; phasor diagram; power flow; effect of varying excitation (V-curves and inverted V-curves); synchronous condenser; applications.	Approx. 14 hours	CO3	Module 3
4	Single-Phase and Special Machines	Single-phase induction motors (double revolving field theory); types (split-phase, capacitor-start, capacitor-run, shaded pole); working principle and applications; special machines (universal motor, stepper motor).	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Three-Phase Induction Motors

Constructional details (squirrel cage and slip ring); production of rotating magnetic field; working principle; slip and its significance; torque-slip characteristics; equivalent circuit; starting and speed control; braking.

CO1 · Approx. 16 hours

Synchronous Generators (Alternators)

Constructional features (salient pole and cylindrical rotor); EMF equation; winding factors; armature reaction; synchronous reactance; phasor diagram; voltage regulation (EMF, MMF methods); parallel operation.

CO2 · Approx. 16 hours

Synchronous Motors

Working principle (magnetic locking); starting methods; phasor diagram; power flow; effect of varying excitation (V-curves and inverted V-curves); synchronous condenser; applications.

CO3 · Approx. 14 hours

Single-Phase and Special Machines

Single-phase induction motors (double revolving field theory); types (split-phase, capacitor-start, capacitor-run, shaded pole); working principle and applications; special machines (universal motor, stepper motor).

CO4 · Approx. 14 hours

MODULE 1

Three-Phase Induction Motors

Constructional details (squirrel cage and slip ring); production of rotating magnetic field; working principle; slip and its significance; torque-slip characteristics; equivalent circuit; starting and speed control; braking.

Approx. 16 hours **CO1** **Understand · Apply**

Learning goals

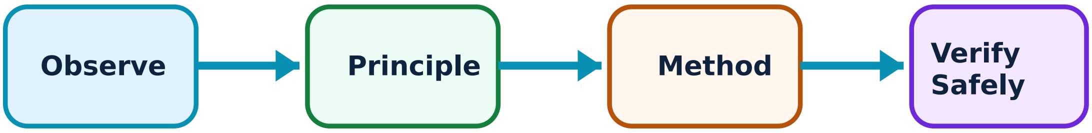
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Three-Phase Induction Motors: identify the system, state its principle, follow the correct method and verify the result safely.

1. Working principle and characteristics−

Three-phase induction motors operate on the principle of electromagnetic induction. A rotating magnetic field (RMF) is produced in the stator, which induces currents in the rotor, creating torque. The torque-slip characteristic shows how torque varies from standstill to synchronous speed.

Study and application method

Sketch the torque-slip characteristic of a 3-phase induction motor and label the motoring, generating and braking regions; explain the concept of 'slip'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the torque-slip characteristic of a 3-phase induction motor and label the motoring, generating and braking regions; explain the concept of 'slip'.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: [illegible] concept [illegible] [illegible] [illegible]. [illegible] diagram / circuit / formula / procedure [illegible] [illegible]. [illegible] result [illegible] application [illegible] [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Common mistake / precaution: Induction motors draw high starting currents. Do not start large motors without authorised starting equipment (e.g., Star-Delta or Auto-transformer starters).

2. Starting and speed control−

Starting methods (DOL, Star-Delta, Auto-transformer, Rotor resistance) limit starting current. Speed control is achieved by varying supply frequency (V/f control), number of poles, or rotor resistance (for slip-ring motors).

Study and application method

Compare the Star-Delta starter with the DOL starter in terms of starting current and torque; describe the principle of V/f speed control.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare the Star-Delta starter with the DOL starter in terms of starting current and torque; describe the principle of V/f speed control.

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Common mistake / precaution: Speed control through frequency variation (VFD) must maintain the V/f ratio to prevent magnetic saturation. Use only authorised drives and control parameters.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Synchronous Generators (Alternators)

Constructional features (salient pole and cylindrical rotor); EMF equation; winding factors; armature reaction; synchronous reactance; phasor diagram; voltage regulation (EMF, MMF methods); parallel operation.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

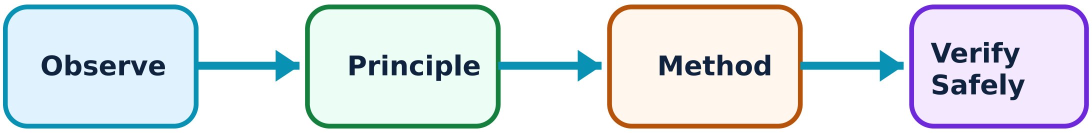
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Synchronous Generators (Alternators): identify the system, state its principle, follow the correct method and verify the result safely.

1. EMF generation and regulation−

Alternators convert mechanical energy into AC electrical energy. The induced EMF depends on flux, speed and winding distribution. Voltage regulation is the change in terminal voltage from no-load to full-load, determined by armature reaction and reactance.

Study and application method

Derive the EMF equation of an alternator considering pitch and distribution factors; explain the 'Synchronous Impedance' method of voltage regulation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Derive the EMF equation of an alternator considering pitch and distribution factors; explain the 'Synchronous Impedance' method of voltage regulation.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Alternators operate at high voltages and speeds. All OC and SC tests for regulation must be conducted under authorised supervision with proper protection.

2. Parallel operation and synchronisation−

Parallel operation increases system reliability and efficiency. Synchronisation requires matching voltage, frequency, phase sequence and phase angle between the alternator and the infinite bus using synchroscopes or lamps.

Study and application method

List the four conditions required for parallel operation of alternators; describe the 'three-lamp' method of synchronisation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List the four conditions required for parallel operation of alternators; describe the 'three-lamp' method of synchronisation.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Incorrect synchronisation can lead to massive mechanical and electrical stresses. Never close the synchronising switch without authorised verification of all conditions.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Synchronous Motors

Working principle (magnetic locking); starting methods; phasor diagram; power flow; effect of varying excitation (V-curves and inverted V-curves); synchronous condenser; applications.

Approx. 14 hours | CO3 | Understand · Apply

Learning goals

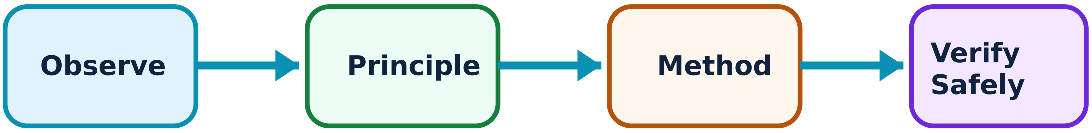
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Synchronous Motors: identify the system, state its principle, follow the correct method and verify the result safely.

1. Working principle and starting−

Synchronous motors run at a constant speed (synchronous speed) regardless of load. They are not self-starting and require external means (damper windings or pony motors) to reach synchronous speed, where magnetic locking occurs between stator and rotor fields.

Study and application method

Explain why a synchronous motor is not self-starting; describe the role of 'damper windings' in starting and hunting prevention.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain why a synchronous motor is not self-starting; describe the role of 'damper windings' in starting and hunting prevention.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Synchronous motors can 'hunt' or lose synchronism under sudden load changes. Do not exceed the 'pull-out torque' specified by the manufacturer.

2. Excitation and V-curves−

Varying the DC field excitation changes the armature current and power factor. V-curves show the relationship between armature current and field current. An over-excited synchronous motor acts as a 'synchronous condenser' for power factor correction.

Study and application method

Sketch the V-curves and inverted V-curves of a synchronous motor; explain how a synchronous motor can improve the power factor of a plant.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the V-curves and inverted V-curves of a synchronous motor; explain how a synchronous motor can improve the power factor of a plant.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Excitation control must be handled by authorised personnel. Excessive field current can lead to rotor overheating and insulation failure.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Single-Phase and Special Machines

Single-phase induction motors (double revolving field theory); types (split-phase, capacitor-start, capacitor-run, shaded pole); working principle and applications; special machines (universal motor, stepper motor).

Approx. 14 hours | **CO4** | **Understand · Apply**

Learning goals

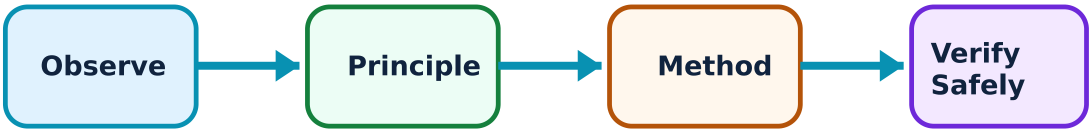
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Single-Phase and Special Machines: identify the system, state its principle, follow the correct method and verify the result safely.

1. Single-phase induction motors−

Single-phase motors are widely used in domestic applications. The double revolving field theory explains why they are not self-starting. Starting is achieved by creating a phase difference between two windings (split-phase or capacitor methods).

Study and application method

Explain the 'Double Revolving Field Theory'; compare the starting torque of a capacitor-start motor with a shaded-pole motor.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the 'Double Revolving Field Theory'; compare the starting torque of a capacitor-start motor with a shaded-pole motor.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: Capacitor-start motors involve high voltages across the capacitor. Do not handle capacitors without authorised discharge procedures.

2. Special electrical machines−

Universal motors operate on both AC and DC, providing high starting torque for appliances. Stepper motors provide precise incremental motion for control systems. These machines are selected based on specialized torque, speed or position requirements.

Study and application method

Describe the construction and applications of a Universal motor; explain the working principle of a permanent-magnet Stepper motor.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the construction and applications of a Universal motor; explain the working principle of a permanent-magnet Stepper motor.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Special machines often require dedicated electronic controllers. Use only authorised drive circuits and power supplies as per machine specifications.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Three-Phase Induction Motors

Use the concept flow to revise observation, principle, method and verification.

Module 2: Synchronous Generators (Alternators)

Use the concept flow to revise observation, principle, method and verification.

Module 3: Synchronous Motors

Use the concept flow to revise observation, principle, method and verification.

Module 4: Single-Phase and Special Machines

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare squirrel cage and slip-ring induction motors in a conceptual table showing construction and applications.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch the phasor diagram of a synchronous generator for lagging, leading and unity power factor loads.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw the V-curves for a synchronous motor and identify the regions of under-excitation and over-excitation.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify the type of single-phase motor used in three different household appliances (e.g., fan, refrigerator, mixer).

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the synchronous speed and slip conceptually for an induction motor given the number of poles and frequency.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure		
Component	**Official mark / conversion**	**Student evidence**
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Three-Phase Induction Motors

Explain Working principle and characteristics and state one correct method, application or precaution.−

Three-phase induction motors operate on the principle of electromagnetic induction. A rotating magnetic field (RMF) is produced in the stator, which induces currents in the rotor, creating torque. The torque-slip characteristic shows how torque varies from standstill to synchronous speed.

Answer framework: Sketch the torque-slip characteristic of a 3-phase induction motor and label the motoring, generating and braking regions; explain the concept of 'slip'.

Important point: Induction motors draw high starting currents. Do not start large motors without authorised starting equipment (e.g., Star-Delta or Auto-transformer starters).

Explain Starting and speed control and state one correct method, application or precaution.−

Starting methods (DOL, Star-Delta, Auto-transformer, Rotor resistance) limit starting current. Speed control is achieved by varying supply frequency (V/f control), number of poles, or rotor resistance (for slip-ring motors).

Answer framework: Compare the Star-Delta starter with the DOL starter in terms of starting current and torque; describe the principle of V/f speed control.

Important point: Speed control through frequency variation (VFD) must maintain the V/f ratio to prevent magnetic saturation. Use only authorised drives and control parameters.

Module 2 — Synchronous Generators (Alternators)

Explain EMF generation and regulation and state one correct method, application or precaution.−

Alternators convert mechanical energy into AC electrical energy. The induced EMF depends on flux, speed and winding distribution. Voltage regulation is the change in terminal voltage from no-load to full-load, determined by armature reaction and reactance.

Answer framework: Derive the EMF equation of an alternator considering pitch and distribution factors; explain the 'Synchronous Impedance' method of voltage regulation.

Important point: Alternators operate at high voltages and speeds. All OC and SC tests for regulation must be conducted under authorised supervision with proper protection.

Explain Parallel operation and synchronisation and state one correct method, application or precaution.−

Parallel operation increases system reliability and efficiency. Synchronisation requires matching voltage, frequency, phase sequence and phase angle between the alternator and the infinite bus using synchroscopes or lamps.

Answer framework: List the four conditions required for parallel operation of alternators; describe the 'three-lamp' method of synchronisation.

Important point: Incorrect synchronisation can lead to massive mechanical and electrical stresses. Never close the synchronising switch without authorised verification of all conditions.

Module 3 — Synchronous Motors

Explain Working principle and starting and state one correct method, application or precaution.−

Synchronous motors run at a constant speed (synchronous speed) regardless of load. They are not self-starting and require external means (damper windings or pony motors) to reach synchronous speed, where magnetic locking occurs between stator and rotor fields.

Answer framework: Explain why a synchronous motor is not self-starting; describe the role of 'damper windings' in starting and hunting prevention.

Important point: Synchronous motors can 'hunt' or lose synchronism under sudden load changes. Do not exceed the 'pull-out torque' specified by the manufacturer.

Explain Excitation and V-curves and state one correct method, application or precaution.−

Varying the DC field excitation changes the armature current and power factor. V-curves show the relationship between armature current and field current. An over-excited synchronous motor acts as a 'synchronous condenser' for power factor correction.

Answer framework: Sketch the V-curves and inverted V-curves of a synchronous motor; explain how a synchronous motor can improve the power factor of a plant.

Important point: Excitation control must be handled by authorised personnel. Excessive field current can lead to rotor overheating and insulation failure.

Module 4 — Single-Phase and Special Machines

Explain Single-phase induction motors and state one correct method, application or precaution.−

Single-phase motors are widely used in domestic applications. The double revolving field theory explains why they are not self-starting. Starting is achieved by creating a phase difference between two windings (split-phase or capacitor methods).

Answer framework: Explain the 'Double Revolving Field Theory'; compare the starting torque of a capacitor-start motor with a shaded-pole motor.

Important point: Capacitor-start motors involve high voltages across the capacitor. Do not handle capacitors without authorised discharge procedures.

Explain Special electrical machines and state one correct method, application or precaution.−

Universal motors operate on both AC and DC, providing high starting torque for appliances. Stepper motors provide precise incremental motion for control systems. These machines are selected based on specialized torque, speed or position requirements.

Answer framework: Describe the construction and applications of a Universal motor; explain the working principle of a permanent-magnet Stepper motor.

Important point: Special machines often require dedicated electronic controllers. Use only authorised drive circuits and power supplies as per machine specifications.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Three-Phase Induction Motors.

2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Synchronous Generators (Alternators).
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Synchronous Motors.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Single-Phase and Special Machines.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Constructional details (squirrel cage and slip ring); production of rotating magnetic field; working principle; slip and its significance; torque-slip characteristics; equivalent circuit; starting and speed control; braking..
2. Develop a complete answer or practical plan for: Constructional features (salient pole and cylindrical rotor); EMF equation; winding factors; armature reaction; synchronous reactance; phasor diagram; voltage regulation (EMF, MMF methods); parallel operation..
3. Develop a complete answer or practical plan for: Working principle (magnetic locking); starting methods; phasor diagram; power flow; effect of varying excitation (V-curves and inverted V-curves); synchronous condenser; applications..
4. Develop a complete answer or practical plan for: Single-phase induction motors (double revolving field theory); types (split-phase, capacitor-start, capacitor-run, shaded pole); working principle and applications; special machines (universal motor, stepper motor)..

LAST-MILE REVISION

Quick Revision

Module 1: Three-Phase Induction Motors

Constructional details (squirrel cage and slip ring); production of rotating magnetic field; working principle; slip and its significance; torque-slip characteristics; equivalent circuit; starting and speed control; braking.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Synchronous Generators (Alternators)

Constructional features (salient pole and cylindrical rotor); EMF equation; winding factors; armature reaction; synchronous reactance; phasor diagram; voltage regulation (EMF, MMF methods); parallel operation.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Synchronous Motors

Working principle (magnetic locking); starting methods; phasor diagram; power flow; effect of varying excitation (V-curves and inverted V-curves); synchronous condenser; applications.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Single-Phase and Special Machines

Single-phase induction motors (double revolving field theory); types (split-phase, capacitor-start, capacitor-run, shaded pole); working principle and applications; special machines (universal motor, stepper motor).

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus

Term / topic	One-line meaning
Working principle and characteristics	Three-phase induction motors operate on the principle of electromagnetic induction.
Starting and speed control	Starting methods (DOL, Star-Delta, Auto-transformer, Rotor resistance) limit starting current.
EMF generation and regulation	Alternators convert mechanical energy into AC electrical energy.
Parallel operation and synchronisation	Parallel operation increases system reliability and efficiency.
Working principle and starting	Synchronous motors run at a constant speed (synchronous speed) regardless of load.
Excitation and V-curves	Varying the DC field excitation changes the armature current and power factor.
Single-phase induction motors	Single-phase motors are widely used in domestic applications.
Special electrical machines	Universal motors operate on both AC and DC, providing high starting torque for appliances.

LEARNING RESOURCES

References

Recommended electrical-machines references

1. B. L. Theraja and A. K. Theraja, A Textbook of Electrical Technology (Vol. II).
2. P. S. Bimbhra, Electrical Machinery.
3. D. P. Kothari and I. J. Nagrath, Electric Machines.
4. M. G. Say, Performance and Design of A.C. Machines.

Approved public electrical-machines sources

- https://nptel.ac.in/courses/108105131
- https://onlinecourses.nptel.ac.in/noc22_ee06/preview

These sources provide the verified study basis while official Course 5031 detail is unavailable; they do not replace official equipment manuals, authorised machine testing, certified designs or authorised grid-connection protocols.